

GRID-CLOCK QUANTUM GRAVITY

Finite Information, Emergent General Relativity, Regular Black Holes, and Unitary Evaporation

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SCOPE NOTE. This paper presents the gravity construction as one continuous candidate theory: finite microscopic information and capacity, finite-Weyl quantum kinematics, strict-local quantum-cellular-automaton propagation, exact microscopic update composition, emergent ADM/DeWitt gravity, finite-capacity strong gravity, regular black-hole cores, horizon boundary information, and unitary evaporation. Statements are explicitly separated into postulates, derived consequences, conditional completion assumptions, numerical or algebraic audits, semiclassical matching results, and genuinely open microscopic problems.

Abstract

Grid-Clock Quantum Gravity (GCQG) is a finite-information candidate theory in which a fixed relational graph and a universal microscopic update index are more fundamental than continuum spacetime. Each elementary capacity cell has bounded local energy, while finite information is represented separately by finite-dimensional Weyl records. Exactness is assigned to finite microscopic update composition rather than to the continuum infinitesimal hypersurface-deformation algebra at fixed cutoff; the ADM/DeWitt algebra is required to emerge in the smooth long-wavelength sector. Geometry-dependent normal deformation generates the inverse-metric H-H structure function, and numerical refinement audits support the H-D and D-D sectors. In the strong-gravity sector, Planck capacity gives a finite density bound and a core radius with cube-root mass scaling. A two-channel Compute-Link closure and a finite-Weyl chord-load completion then yield a conditional Hayward-type regular profile. The present revision adds a strict one-neighborhood-per-update BCC quantum-cellular-automaton propagation candidate. In the minimal linear isotropic Weyl class, published classifications support the BCC/Weyl structure, while a doubled four-component Clifford carrier supplies two positive-frequency propagation amplitudes. The local normal-current projector has rank two for every orientation, so the effective outgoing boundary algebra is two-dimensional. Under Boundary Record Minimality this selects a binary horizon record rather than assuming a qubit by hand. Gauge-reduced boundary counting and semiclassical horizon thermodynamics then fix the corresponding entropy and area increment. The resulting discrete horizon tower admits an explicit energy-conserving unitary evaporation map, yields the microcanonical detailed-balance factor from state counting, and produces the standard Page turnover in the macroscopic limit. Causality and collective horizon updating reproduce the Hawking mass scalings for emission rate, power, and lifetime. A horizon-blindness argument removes the need for an independent phenomenological surface coupling: microcanonical reciprocity relates inclusive spontaneous emission to the inverse absorption problem. The remaining tasks are computational and structural—chiefly the full nonlinear geometry/matter QCA, the QCA greybody spectrum, exclusive microstate amplitudes and level splittings, and a derivation of Boundary Record Minimality. The framework is therefore a sharpened but still conditional finite-information quantum-gravity completion with regular black holes and unitary information transfer.

Keywords: quantum gravity; finite information; finite Weyl algebra; quantum cellular automaton; BCC lattice; emergent spacetime; ADM/DeWitt gravity; regular black holes; horizon entropy; unitary evaporation; Page curve

Claim-status summary

Element	Status	Role
Fixed graph and universal update index	Foundational postulates	Define the microscopic substrate.
Finite local Hilbert space	Additional quantum postulate	Finite information is explicit.
Finite local energy/capacity	Foundational completion postulate	Sets the ultraviolet density scale.
Finite-Weyl algebra	Preferred microscopic carrier	Exact finite unitary record/shift algebra.
Hard projection	Negative-control diagnostic	Shows boundary leakage/operator proliferation.
Exact microscopic update composition	Candidate exact ultraviolet principle	Closure belongs to the finite update/groupoid level.

Element	Status	Role
ADM/DeWitt HDA	Infrared universality target	Recovered in smooth/refinement sectors.
Exact full-cutoff infinitesimal HDA	Rejected in current site realization	Blocked by finite-commutative derivation no-go.
Planck-density core scaling	Robust derived result	Gives cube-root core-radius scaling with mass.
Quadratic load law and regular profile	Conditional completion	Requires the stated two-channel/chord-load assumptions.
Horizon-Link entropy	Candidate boundary algebra + IR matching	Rank-two boundary algebra selects a binary boundary record; the area-law coefficient follows conditionally from semiclassical micro-IR matching and Boundary Record Minimality.
Full many-body Lorentzian completion	Open	Strict-local propagation is explicit at candidate level; the full nonlinear geometry/matter QCA and physical-state construction remain open.

1. Introduction

The central difficulty of quantum gravity is structural. General relativity is not merely a quantum field theory waiting to be placed on a fixed background: its Hamiltonian formulation is constrained, the spatial metric is dynamical, and the brackets of the Hamiltonian and spatial-diffeomorphism constraints contain the inverse metric as a structure function. A microscopic theory must therefore explain both ultraviolet finiteness and the emergence of the general-relativistic deformation structure.

The present framework reverses the usual order of construction. It does not begin by quantizing a continuum metric. It begins with a fixed graph of adjacency relations and a universal ordering of microscopic updates. Proper time, physical distance, smooth geometry, and continuum constraints are collective descriptions of a finite-information substrate. The theory then asks four linked questions: what finite quantum algebra should live on the graph; what notion of closure should be exact; how smooth general relativity can emerge; and what happens when local gravitational load approaches the microscopic capacity.

$$\begin{aligned}
 & \textit{finite information} \rightarrow \textit{finite quantum update algebra} \rightarrow \textit{emergent geometry} \\
 & \rightarrow \textit{ADM/DeWitt gravity} \rightarrow \textit{regular strong gravity}.
 \end{aligned}$$

The remainder of this paper develops that chain in detail. Where an assumption is additional rather than derived, it is identified explicitly. Numerical results are treated as pilot consistency tests rather than as proofs.

PART I. MICROSCOPIC FINITE-INFORMATION ARCHITECTURE

2. Capacity, information, and logical status

Finite energy capacity

$$\begin{aligned}
 E_{cell} &\leq E_p \\
 \rho_p &= \frac{m_p}{\ell_p^3}
 \end{aligned}$$

The energy-capacity postulate fixes the maximum local energy or density scale once a microscopic cell realization is specified. It motivates a Planck-density central cap but does not mathematically imply a finite-dimensional Hilbert space.

Finite Local Information Postulate

$$\dim(H_{cell}) = d < \infty$$

Finite local information is therefore retained as an additional explicit quantum postulate. A bounded Hamiltonian may exist on an infinite-dimensional Hilbert space, so finite energy and finite information must remain logically distinct assumptions.

Bounded observables and records

$$\eta = H_{cell}/E_p, 0 \leq \eta \leq 1$$

Physical storage, geometry, and Link observables are represented by bounded Hermitian functions of finite records. The finite-Weyl shift itself is not interpreted as an energy-raising operator; its cyclicity is an information-space property.

3. Diagnostic ultraviolet model: truncation and hard projection

Finite-edge diagnostic

$$[a, a^\dagger] = I - d\Pi_{sat}, \Pi_{sat} = |d-1\rangle\langle d-1|$$

A truncated oscillator demonstrates that exact Heisenberg commutation cannot hold globally in finite dimension. Low states can approximate the ordinary oscillator, while the spectral edge necessarily deforms it. This remains a useful diagnostic but not the preferred microscopic GCC algebra.

Exact projected-commutator identity

$$\begin{aligned} P &= P_{cap}, Q = I - P, H^-[N] = PH[N]P \\ [PAP, PBP] &= P[A, B]P - PAQB P + PBQAP \\ \Delta_{cap}[N, M] &= PH[M]QH[N]P - PH[N]QH[M]P \end{aligned}$$

The identity is exact and exposes capacity-boundary leakage. Exploratory closure audits showed, however, that repeated commutators can generate a large saturation-conditioned operator sector. Hard projection is therefore retained as a negative-control completion rather than the preferred ultraviolet architecture.

4. Preferred microscopic carrier: finite Weyl and symplectic updates

Finite Weyl algebra

$$\begin{aligned} Z|n\rangle &= \omega^n|n\rangle \\ Xe_n &= e_{(n+1) \bmod d} \\ \omega &= \exp\left(\frac{2\pi i}{d}\right) \\ ZX &= \omega XZ \end{aligned}$$

The pair Z,X is an exact finite unitary clock/shift algebra. A phase-symmetrized Weyl basis W(u) closes under multiplication, and its commutator has bounded trigonometric structure coefficients. For low phase-space labels the sine factor is linear and ordinary canonical brackets are recovered as a smooth approximation.

Finite symplectic/Clifford sector

$$G_{micro} \sim HW_d^N \rtimes Sp(2N, Z_d)$$

Finite symplectic transformations preserve the Weyl symplectic product and are implemented by finite Clifford unitaries in the standard admissible cases. This gives an exactly closed finite update carrier. The notation is architectural: details depend on whether d is prime, prime-power, or a more general ring-valued register.

Compute-Link interpretation

Local phase operations represent Compute/record refreshes, while relative Weyl products between neighboring cells represent Link records. Because these are finite unitaries from the outset, no spectral-edge projector is required to define the microscopic state space.

5. Compute-Link quantum factorization and correlations

$$H_{cell} = H_C \otimes H_L$$

$$\chi_c = \text{Tr}(\rho P_c), \chi_L = \text{Tr}(\rho P_L), q_{CL} = \text{Tr}(\rho P_c P_L)$$

$$\rho/\rho_p = q_{CL}$$

For an uncorrelated Compute-Link state, the joint support factorizes. Under the minimum isotropic closure the two sector loads are equal, so

$$\rho/\rho_p = \chi^2$$

For a correlated state define

$$C_{CL} = \langle P_c P_L \rangle - \langle P_c \rangle \langle P_L \rangle$$

$$\frac{\rho}{\rho_p} = \chi^2 + C_{CL}$$

Thus the quadratic regular-core law is the uncorrelated semiclassical limit of a two-sector quantum architecture. The new Weyl Link construction below supplies a second, complementary interpretation of the same quadratic response.

5.1 Explicit reversible Compute-Link correlation circuit

The covariance term need not be left as an arbitrary radial function. A minimal reversible dilation uses effective two-level Compute and Link records, a memory register that stores the previous Link state, and a scheduler qubit. The input state is diagonal in the local record basis, and a first-harmonic Link phase drives the scheduler rotation. The resulting branch probability is

$$\rho_\chi = (1 - \chi)|0\rangle\langle 0| + \chi|1\rangle\langle 1|, \quad \rho_{in} = \rho_\chi^{(C)} \otimes \rho_\chi^{(L)}, \quad \Phi = 2g\chi^2, \quad \lambda(\chi) = \sin^2(g\chi^2)$$

On the correlated branch, apply a Compute-to-Link controlled-NOT and then swap the Link record with the memory register. After tracing the reversible work registers, the Compute-Link support becomes

$$q_{CL} = (1 - \lambda)\chi^2 + \lambda\chi$$

$$C_{CL} = \chi(1 - \lambda)\sin^2(g\chi^2)$$

$$\frac{\rho}{\rho_p} = f_g(\chi) = \chi^2 + \chi(1 - \lambda)\sin^2(g\chi^2)$$

The correction is bounded, vanishes at both endpoints, and first appears at fifth order in the weak-load expansion. It therefore leaves the leading exterior correction unchanged and contributes only at substantially higher inverse powers of radius. The Link-capacity allocation further imposes the bound

$$g \leq \frac{f_L}{z}$$

6. Quantum metric records and dynamic geometry

Bounded metric record

$$h^{ab} = (EE^T)^{ab}$$

A finite geometry register may encode an inverse-triad-like record. The metric is then defined through the positive construction shown above. Quantum superpositions of finite record labels produce operator-valued metric records without requiring the microscopic coordinate labels themselves to be continuous.

Conjugate geometry refresh

Exact finite-dimensional canonical commutation is impossible globally. The conjugate geometry refresh is therefore represented by a finite Weyl shift/phase pair. In a low-phase chart the finite pair approximates canonical variables, while the exact microscopic evolution remains finite and unitary.

Geometry-controlled update

$$U_N = \sum_g |g\rangle\langle g| \otimes U_N[g]$$

A finite geometry state can control the local symplectic/Clifford scheduler. Geometry and matter then entangle under a unitary update. In a semiclassical state sharply peaked on one record, the conditional evolution reduces to the corresponding classical metric sector.

7. Infrared constraint algebra and ADM/DeWitt target

Hypersurface-deformation target

$$[H[N], H[M]]/(\hbar) \rightarrow D[\omega]$$

$$\omega^a = \hbar^{ab} (N \partial_b M - M \partial_b N)$$

The fixed-graph reconstruction developed here obtains this structure from record redundancy, smooth address gauge, and scheduler non-observability. The finite carrier is required to reproduce that algebra in the low-mode refinement limit.

DeWitt kinetic sector

$$H_\lambda[N] = \int d^3x N \left[\frac{\pi^{ab} \pi_{ab} - \lambda \pi^2}{\sqrt{\hbar}} - \sqrt{\hbar} R^{(3)} \right]$$

In three spatial dimensions the general-relativistic DeWitt coefficient takes its standard value in this convention. Independent refinement audits in the infrared reconstruction found closure roots converging to that value and regulator anomalies decreasing with spatial refinement. The finite-Weyl carrier is not used to alter this infrared target.

Full closure requirement

A viable ultraviolet completion must respect H-H, H-D, and D-D together. Bracket-only deformations are insufficient: a state-dependent scalar factor or a componentwise bounded-momentum substitution can look acceptable in H-H and still generate anomalies in the remaining brackets.

8. Closure by automorphism and linear microscopic selection

Finite-Weyl Closure Principle

Let the finite-unitary conjugation map act on observables in the usual way. Then

$$[A_U(A), A_U(B)] = A_U([A, B])$$

Hence a Planck-scale finite-information deformation is anomaly-free at the operator-algebra level if it acts on the complete constraint representation as an automorphism rather than modifying one Hamiltonian term in isolation. This does not by itself create new physics; physical record maps must also be fixed, otherwise a global conjugation can be a basis change.

Physical-record-preserving linear maps

Take finite phase-space labels and a general linear symplectic map. If the equal-update physical record is left unchanged by an internal Link refresh, the symplectic conditions reduce the map to

$$q' = q, p' = p + Cq, C = C^T$$

Fixed-graph locality sets non-neighbor couplings to zero, while uniform-vacuum invariance annihilates the constant mode. These conditions identify the coupling matrix with a weighted graph Laplacian; in a homogeneous isotropic reference sector the minimal choice is the graph Laplacian scaled by the scheduler parameter.

Reversible scheduler and discrete wave equation

Combining the record-preserving shear with a finite Fourier/Clifford exchange gives the symplectic recurrence

$$q_{K+1} - 2q_K + q_{K-1} = -\kappa L_G q_K$$

Equivalently, using the opposite-sign graph-Laplacian convention, the update is the GCC discrete wave equation. For a Laplacian eigenmode, the dispersion relation is

$$\cos(\omega) = 1 - \kappa\lambda/2$$

In a nearest-neighbor one-dimensional reference chain, the Laplacian spectrum has the standard sine-squared form. With the one-link-per-tick calibration, the principal branch is linear and the real-space solution decomposes into

left- and right-moving records. Thus the minimum reversible finite scheduler reproduces exact one-link-per-tick propagation in this benchmark.

9. Nonlinear finite-Weyl Link completion

General local harmonic form

Let each finite record carry a phase label and define the relative Link phase between neighboring cells. Locality, exchange symmetry, invariance under a common phase shift, and a zero-energy uniform record imply that a real Link energy can be expanded as

$$V(\delta) = \sum_{m \geq 1} a_m [1 - \cos(m\delta)]$$

The infrared quadratic coefficient fixes only one weighted sum over the harmonic amplitudes. Therefore finite information, locality, and the infrared wave equation alone do not uniquely select a nonlinear ultraviolet interaction.

Weyl chord load

Define the normalized chord load of two unit-circle Weyl records by

$$\chi_W = |\sin(\delta/2)|, 0 \leq \chi_W \leq 1$$

This is a completion dictionary between relative finite-Weyl geometry and a dimensionless local Link load. It is geometrically natural but is not derived from the original GCC postulates. For the primitive first harmonic,

$$1 - \cos(\delta) = 2\chi_W^2$$

so the nonlinear periodic interaction is exactly quadratic in the chord load even at saturation.

Positive-harmonic uniqueness proposition

Assume nonnegative harmonic coefficients, normalize the fully separated record to the saturation scale, and normalize the response by its saturated value. Then

$$V(\pi) = 2 \sum_{m \text{ odd}} a_m$$

Near the vacuum, the relative phase is twice the chord load to leading order, so

$$f(\chi_W) = \chi_W^2 \frac{\sum m^2 a_m}{\sum_{m \text{ odd}} a_m} + O(\chi_W^4)$$

Requiring unit quadratic normalization in the infrared therefore gives

$$\sum_m m^2 a_m = \sum_{m \text{ odd}} a_m$$

For nonnegative coefficients, all even harmonics contribute only positive terms to the difference between the two sides, and all odd harmonics above the fundamental add a positive excess. Equality therefore forces every coefficient except the first harmonic to vanish. Within this stated positive-harmonic sector, the first harmonic is unique up to the overall energy scale:

$$V_{link}(\delta) = \epsilon_L [1 - \cos \delta] = 2\epsilon_L \chi_W^2$$

The proposition fails to be unique if negative harmonic coefficients are permitted and fine-tuned cancellations are allowed. Positivity is therefore a substantive completion condition, not a mathematical consequence of finite Weyl kinematics alone.

Bounded Link response

$$0 \leq V_{link} \leq 2\epsilon_L$$

The Link energy and its finite phase kick are bounded. If the graph has coordination number z and Link energy is shared between incident cells, the Link share of the cell-capacity budget obeys the following bound. A fractional allocation simply rescales the right-hand side.

$$z\epsilon_L \leq E_P, \quad z\epsilon_L = f_L E_P$$

PART II. MICROSCOPIC CLOSURE AND EMERGENT GENERAL RELATIVITY

10. Why exact finite-cutoff infinitesimal HDA is too strong

A finite-commutative derivation no-go

Let the microscopic site algebra be the algebra of complex-valued functions on the finite graph with pointwise multiplication, and let a derivation satisfy the ordinary Leibniz rule. For the indicator basis, idempotence gives

$$\delta(e_i) = \delta(e_i^2) = 2e_i\delta(e_i)$$

Evaluating the identity at the selected site forces that component to vanish; every other component vanishes because the indicator itself is zero there. Thus each basis idempotent is annihilated, and therefore

$$\delta = 0.$$

Thus a finite commutative algebra of site values admits no nonzero derivation with the ordinary continuum Leibniz property. This does not forbid finite difference operators, noncommutative differential calculi, or exact finite group actions. It forbids identifying an ordinary continuum spatial derivative with a nontrivial exact derivation of the finite pointwise site algebra.

Consequence for the closure target

The conclusion is not that discrete gravity cannot possess exact symmetry. Rather, exactness should be formulated at the level appropriate to the microscopic variables: finite transformations, noncommutative record algebras, groupoids, or specially constructed perfect discretizations. The continuum HDA can then be a derived low-energy algebra. This distinction parallels the general observation that discretization typically breaks continuum diffeomorphism symmetry, while specially improved or perfect constructions can restore exact symmetry in suitable sectors.

11. Exact microscopic Weyl composition

To verify that finite microscopic exactness is not merely formal, Eq. (3.1) was evaluated as an explicit matrix identity. For each d , X and Z were constructed as $d \times d$ unitary matrices and the relative Frobenius residual was computed:

$$R_W = | |W_M W_N - \omega^k W_N W_M| |_F / | |W_M W_N| |_F$$

The residuals are at floating-point roundoff. The finite Weyl sector therefore provides a concrete exactly closed microscopic operator algebra at finite Hilbert-space dimension.

12. Geometry-derived normal-normal bracket

Continuum geometric identity

Consider a spatial hypersurface with an embedding, tangent vectors, induced metric, and unit normal. A normal deformation with lapse N is

$$\delta_N X = Nn.$$

Varying the orthogonality and unit-normalization conditions gives a purely tangential normal variation. Because the tangent vector changes by differentiating the normal displacement,

$$(\delta_N n) \cdot e_a = -\partial_a N$$

Writing the normal variation in the tangent basis and solving with the inverse induced metric gives

$$\delta_N n = -h^{ab}(\partial_b N)e_a.$$

The order difference of two normal deformations is therefore

$$[\delta_{N'}, \delta_M]X = e_a h^{ab}(N\partial_b M - M\partial_b N).$$

Equation (6.3) is the H-H hypersurface-deformation structure function. Importantly, the inverse metric arises from solving the orthogonality condition for the updated normal; it need not be inserted independently into the primitive normal-refresh rule.

Fixed-graph discretization

On a graph refinement of a smooth patch, tangent records are reconstructed locally from neighboring embedding records. A minimal centered stencil reconstructs each tangent from the two neighboring records. The normal is then obtained as the normalized vector orthogonal to the three local tangents. After the first normal refresh, all tangents and normals are recomputed from the updated records before the second refresh. This state dependence is essential; reusing the original normal would remove the metric-dependent response.

Three-dimensional non-orthogonal pilot audit

A reproducible pilot calculation used a cubical graph embedded in four dimensions. The embedding deformation and the two lapse profiles were

$$\begin{aligned} F &= 0.18uv + 0.11uw - 0.14vw + 0.07u^2 - 0.05v^2 + 0.04\sin(\pi u)\sin(\pi w) \\ N &= 0.75 + 0.12u - 0.08v + 0.06w + 0.04uv \\ M &= 0.65 - 0.09u + 0.11v + 0.05w - 0.03vw \end{aligned}$$

For each deformation size, the two finite update paths were computed with the normal recalculated after the first step. Their order difference was fitted to the tangential vector predicted by Eq. (6.3), with the expected quadratic dependence on the step size. Two outer graph layers were excluded from the norm.

A log-log fit gives the quoted raw and residual scaling exponents. At the representative small-step value, using the full inverse metric reduces the relative vector residual by more than an order of magnitude compared with replacing it by the identity. This is a pilot consistency check, not a proof for arbitrary graphs, but it verifies the required order structure and the necessity of the inverse-metric response in a non-orthogonal three-dimensional example.

13. Full infrared hypersurface-deformation target

In the smooth limit the effective normal and tangential generators are required to satisfy the standard geometric deformation relations, up to conventional signs:

$$\begin{aligned} [\delta_N, \delta_M] &= \delta_v, v^a = h^{ab}(N\partial_b M - M\partial_b N) \\ [\delta_v, \delta_N] &= \delta_{L_v N} \\ [\delta_v, \delta_w] &= \delta_{[v,w]} \end{aligned}$$

At the canonical level these become the H-H, D-H, and D-D sectors of the ADM/DeWitt constraint algebra. The ADM/DeWitt theory is the required infrared universality class. The remaining question is how the finite graph approaches this target and where exactness can consistently be imposed.

14. Finite-difference closure audit: full cutoff versus smooth sector

Kinematic test operators

To isolate finite-difference effects, consider a periodic one-dimensional graph with a centered derivative. For a site lapse and tangential shift, define the multiplication and transport operators in the standard way. The continuum identities suggest

$$\begin{aligned} [L_v, M_N] &= M_{v\partial N} \\ [L_v, L_w] &= L_{v\partial w - w\partial v} \end{aligned}$$

At finite spacing we compare the left-hand sides with the same expressions using the centered derivative. The smooth-sector restriction keeps only a fixed band of low Fourier modes while the graph is refined.

Full-cutoff operator residual

The full matrix residuals do not decrease toward zero because the highest graph modes remain at the cutoff and the centered difference operator is not an exact derivation of the finite pointwise algebra. This numerical behavior is the finite-difference manifestation of the no-go result in Section 4.

Fixed smooth-sector residual

With the low Fourier sector held fixed while the graph is refined, the residuals scale as

$$A_{HD}^{low} \propto a^{1.987}, \quad A_{DD}^{low} \propto a^{1.981}$$

If both input and output are projected back to the same low sector, the measured exponents are 1.995 and 1.999 respectively. Thus the ordinary infinitesimal deformation algebra is recovered with the expected second-order accuracy of the centered derivative in the smooth sector, even though it is not an exact identity on the complete finite-cutoff operator space.

15. From geometry response to a finite-Weyl scheduler

Equation (6.2) suggests a microscopic dictionary that does not hard-code the continuum HDA. Let the primitive geometry update define a finite record from the measured tangential response of the normal. In the low-load chart this record approaches the inverse-metric-weighted lapse gradient. Quantize the lapse amplitude and this response into finite Weyl labels and use them as exponents.

$$W_N^a = X^{A_N} Z^{B_N^a}$$

The order dependence of two such updates is controlled exactly by the finite symplectic product

$$\kappa^a = A_N B_M^a - A_M B_N^a.$$

After conversion back to a smooth chart, Eq. (9.1) approaches

$$\kappa^a / (s_A s_B) \rightarrow N b_M^a - M b_N^a \rightarrow h^{ab} (N \partial_b M - M \partial_b N)$$

This construction separates two questions. The Weyl composition is exact at finite d. The identification of its finite labels with continuum lapse/normal-response variables is an emergent chart whose accuracy is controlled by graph and record refinement.

16. Spatial address and internal frame gauge

Address relabeling

In the fixed-graph interpretation, a tangential deformation is not a literal motion of the fundamental graph. It is an update of relational address records or a graph relabeling in the smooth description. The compensating finite shift therefore belongs naturally to an address-gauge sector rather than to a new propagating degree of freedom.

Triad redundancy

$$h = EE^T, E \rightarrow ER, R \in SO(3)$$

A local triad record contains a frame redundancy invisible to the metric. This supplies an independent internal gauge sector. Finite $SO(3)$ frame transformations can be lifted to finite-dimensional $SU(2)$ representations. The present paper does not claim that an explicit Gauss term must appear in every H-H bracket; that depends on variables and gauge choice. The key requirement is that internal frame changes remain within the allowed gauge fiber rather than appearing as an anomaly outside the physical record algebra.

17. Two-level closure principle

The combined results motivate the following GCC closure principle.

Microscopic level: $A_{micro} = 0$ for exact finite update/groupoid composition

Emergent level: $A_{eff} \rightarrow 0$ in the smooth low – phase refinement sector

These statements are logically different. The first concerns exact finite quantum consistency. The second concerns recovery of the continuum first-class constraint algebra. Demanding the second as an exact identity on the entire finite-cutoff commuting site algebra is neither necessary for emergence nor compatible with the derivation no-go of Section 4.

This viewpoint also clarifies unitary automorphisms: a complete conjugation preserves the commutator structure of an already closed algebra, but it carries any existing anomaly along with it. The framework therefore places exactness in the primitive finite update law rather than treating a basis change as a closure mechanism.

18. ADM/DeWitt infrared limit

In the infrared, GCC requires canonical metric variables to approach the ADM/DeWitt form. In three spatial dimensions the Hamiltonian density is, in standard conventions,

$$H = \frac{1}{\sqrt{h}} \left(\pi^{ab} \pi_{ab} - \frac{1}{2} \pi^2 \right) - \sqrt{h} R^{(3)}$$

$$H_a = -2D_b \pi_a^b$$

The finite-Weyl composition law alone does not determine the DeWitt kinetic coefficient. The infrared reconstruction separately selects the ADM/DeWitt universality class. The result established here is compatibility of a finite exactly closed microscopic carrier with the geometric structure function and with the full smooth HDA target.

18.1 Reproducible numerical closure-audit tables

d	Weyl phase exponent	Relative matrix residual
5	2	2.18×10^{-16}
7	4	5.75×10^{-16}
11	8	3.54×10^{-16}
17	14	4.31×10^{-16}

epsilon	Raw normal-normal diamond	Residual after fitted tangential shift	Fitted coefficient
0.040	3.17×10^{-4}	2.52×10^{-6}	0.99552
0.020	7.93×10^{-5}	3.01×10^{-7}	0.99723
0.010	1.98×10^{-5}	3.49×10^{-8}	0.99810
0.005	4.96×10^{-6}	4.16×10^{-9}	0.99853

Sites	a	Full-cutoff H-D residual	Full-cutoff D-D residual
32	0.19635	1.2288	1.1354
64	0.09817	1.2257	1.1224
128	0.04909	1.2250	1.1191
256	0.02454	1.2248	1.1183

Sites	a	Low-mode H-D residual	Low-mode D-D residual
32	0.19635	0.11249	0.17904
64	0.09817	0.02875	0.04623
128	0.04909	0.007227	0.01165
256	0.02454	0.001809	0.002918

18.2 Strict-local BCC QCA and processing-scale audit

Primitive QCA rather than a finite-time local Hamiltonian

If strict one-neighborhood-per-update causality is fundamental, the primitive evolution should be a finite-radius unitary QCA rather than the finite-time exponential of a local Hamiltonian. Powers of a nearest-neighbor Hamiltonian generate longer-range tails, so the QCA itself is taken as fundamental and the Hamiltonian only as an infrared effective generator.

$$H_{eff} = \frac{i\hbar}{t_*} \text{Log} U_{QCA}$$

Minimal BCC carrier

A strict-local candidate is obtained on the eight-neighbor BCC graph with a doubled Clifford carrier. Define the directional Clifford matrices, projectors, and one-step coefficients as follows.

$$\alpha_i = \sigma_x \otimes \sigma_i, \quad P_i^\pm = \frac{I \pm \alpha_i}{2}, \quad A_s = P_x^s P_y^s P_z^s, \quad \psi_{K+1}(n) = \sum_s A_s \psi_K(n - s)$$

This is a single radius-one convolution, not three physically sequential substeps. Its support therefore advances by at most one graph neighborhood per universal update. The long-wavelength branch is isotropic and its speed is fixed by the relational spacing per tick. The BCC construction saturates the three-dimensional isotropic speed bound of this directional class. The four-component carrier contains two positive-frequency amplitudes and is the minimum doubled Clifford realization used here for the two physical tensor polarizations.

Published classifications of homogeneous, local, isotropic and unitary two-component quantum walks in three spatial dimensions find a unique admissible Cayley-graph geometry and two Weyl walks, corresponding to the two chiralities [10,11]. The present four-component propagation carrier can be viewed as a doubled-Weyl structure. This supports BCC as the preferred minimal linear propagation graph; it is not a theorem excluding all higher-dimensional, nonlinear or non-Abelian QCA completions.

Relational processing length and capacity consistency

The graph-edge length is not identified directly with the observed continuum distance. Define instead the relational second-moment processing length

$$\ell_{proc}^2 = \frac{1}{3z} \sum_s h_{ab} d_s^a d_s^b$$

For the isotropic BCC neighbor shell, the relational processing length equals the auxiliary lattice scale, and the physical node volume is four times its cube. The static graph Laplacian and the dynamical infrared cone are then normalized by the same relational scale.

Finite energy capacity and finite information dimension must remain separate. In particular, the four-component propagation carrier does not imply four times the energy capacity. The earlier Minimal Action Saturation identification is therefore not adopted here as a derived law. If one BCC processing node is identified with one GCC capacity cell at the Planck energy and Planck density, consistency requires

$$4\ell_{proc}^3 = \ell_p^3$$

$$\ell_{proc} = 4^{-1/3} \ell_p, \quad t_* = 4^{-1/3} t_p$$

This node-to-cell calibration is only one branch. If a QCA node instead coarse-grains several capacity cells, the numerical processing length changes while the separation between energy capacity and information dimension remains mandatory.

PART III. FINITE CAPACITY AND REGULAR STRONG GRAVITY

19. Fundamental cell capacity

Let one elementary grid cell possess a maximum energy capacity equal to the Planck energy:

$$E_{cap} = E_p = m_p c^2.$$

The elementary reference volume is

$$V_p = \ell_p^3.$$

Hence the maximum mass-density scale is

$$\rho_p = m_p / \ell_p^3 = c^5 / (\hbar G^2).$$

The physical interpretation is that no elementary cell is required to represent arbitrarily large localized energy content. Define the cell occupancy fraction

$$\eta_i = E_i / E_p, 0 \leq \eta_i \leq 1.$$

The storage-saturation limit should be distinguished from processing or Link saturation associated with lapse suppression. A black-hole horizon need not consist of Planck-density cells. In the present completion, the finite-density core and the horizon-scale scheduler response remain logically distinct.

20. Core radius from capacity alone

Consider a spherically symmetric object of total mass M . At the limiting density, the smallest volume that can store the mass is fixed by mass divided by the capacity density. For a spherical core,

$$(4\pi/3)r_c^3 = M/\rho_p.$$

Therefore

$$r_c = [3M/(4\pi\rho_p)]^{1/3}.$$

Using Planck quantities,

$$r_c = \ell_p [3M/(4\pi m_p)]^{1/3}.$$

Thus

$$r_c \propto M^{1/3}.$$

This scaling does not depend on a Hayward metric, on the two-channel rule, or on the finite-Weyl completion. It follows directly from finite Planck cell capacity, three-dimensional volume counting, and finite total mass. The numerical coefficient is specific to spherical packing at saturation density; the cube-root mass scaling is the robust part of the argument.

21. Local load variable

Let $m(r)$ denote the mass enclosed inside areal radius r . Define the mean enclosed density

$$\rho^-(r) = 3m(r)/(4\pi r^3).$$

The corresponding dimensionless capacity-load variable is

$$\chi(r) = \rho^-(r)/\rho_p = 3m(r)/(4\pi\rho_p r^3).$$

This variable is constructed from the state enclosed by the local spherical region and does not explicitly reference the final asymptotic mass. Complete saturation corresponds to unit load.

22. Two-channel scheduler completion and microscopic Link bridge

The GCC interpretation distinguishes two operational roles: local state computation (Compute) and transfer/consistency of neighboring state records (Link). Introduce the additional completion hypothesis that stable local M-state support requires both Compute and Link commits.

$$\rho/\rho_p = \chi_c \chi_L.$$

For the minimum spherical and isotropic model, assume that the two channels respond to the same local load variable:

$$\chi_c = \chi_L = \chi.$$

Then

$$\rho(r) = \rho_p \chi(r)^2.$$

Substitution of Eq. (4.2) gives

$$\rho(r) = \rho_p \left[3m(r)/(4\pi\rho_p r^3) \right]^2.$$

Equation (5.3) remains the central completion assumption of the regular-core derivation. It is not derived from the cell-capacity bound alone. Its role is to translate the finite capacity of two required local processing channels into a constitutive relation for the coarse-grained density profile.

Microscopic finite-Weyl bridge

The finite-Weyl Link construction supplies a candidate microscopic realization of the same quadratic response. Let neighboring records carry phases and define their relative phase. The normalized chord load is

$$\chi_w = |\sin(\delta/2)|, 0 \leq \chi_w \leq 1.$$

For the primitive first harmonic of a local, exchange-symmetric finite-Weyl Link energy,

$$V_{link}(\delta) = \epsilon_L [1 - \cos(\delta)] = 2\epsilon_L \chi_W^2$$

If the coarse spherical load is identified with this finite-Weyl chord load,

$$\chi_W(r) = \chi(r) = \frac{3m(r)}{4\pi\rho_p r^3}$$

then the normalized Link response is exactly quadratic at all loads, including saturation. This supplies a microscopic realization of Eq. (5.3), not an independent derivation from the original GCC postulates. The dictionary in Eq. (5.7) is an explicit completion choice.

Conditional first-harmonic selection

A general real, local, shift-invariant and exchange-symmetric periodic Link energy can be written

$$V(\delta) = \sum_{m \geq 1} a_m [1 - \cos(m\delta)]$$

Assume nonnegative harmonic coefficients, normalize complete phase separation to the saturation scale, and require the normalized infrared response to have unit quadratic coefficient. The coefficient condition is

$$\sum_m m^2 a_m = \sum_{m \text{ odd}} a_m$$

For nonnegative coefficients, every even harmonic contributes positively only to the left side, while every odd harmonic above the fundamental contributes a positive excess. Equality therefore forces all higher harmonics to vanish. Within this stated positive-harmonic sector, the first cosine harmonic is unique up to its overall energy scale. Negative coefficients or more general microscopic interactions can evade this proposition, so full microscopic uniqueness remains open.

23. Analytic regular mass profile

Spherical mass conservation gives

$$dm/dr = 4\pi r^2 \rho(r).$$

Using Eq. (5.4),

$$dm/dr = 9m(r)^2 / [4\pi\rho_p r^4].$$

Equivalently,

$$d[1/m(r)]/dr = -9/[4\pi\rho_p r^4].$$

Integration yields

$$1/m(r) = C + 3/[4\pi\rho_p r^3].$$

Imposing the asymptotic mass condition fixes the integration constant, so

$$m(r) = Mr^3 / [r^3 + 3M/(4\pi\rho_p)].$$

Defining

$$a^3 = \frac{3M}{4\pi\rho_p}$$

the result takes the compact form

$$m(r) = Mr^3 / (r^3 + a^3).$$

This is a Hayward-type regular-black-hole mass profile. It follows analytically from the minimal two-channel local finite-capacity closure. The finite-Weyl Link construction supplies a conditional microscopic realization of the same closure.

24. Central and exterior limits

Central limit

For $r \ll a$, Eq. (6.7) gives

$$m(r) \simeq (4\pi/3)\rho_p r^3.$$

Therefore the central density is finite and saturates at the capacity scale:

$$\rho(0) = \rho_p.$$

In the standard static spherical line element, the central behavior is de Sitter-like rather than singular.

Exterior limit

For $r \gg a$,

$$m(r) = M[1 - a^3/r^3 + o(r^{-6})].$$

Consequently the leading correction to the Schwarzschild metric function is

$$\Delta f(r) \propto r^{-4}.$$

The correction is strongly localized toward the deep interior and strong-field region, while the far exterior rapidly approaches general relativity.

25. Fixing the strong-gravity length scale

In geometrized units, write the regular profile in the form

$$m(r) = \frac{Mr^3}{r^3 + 2M\ell_\star^2}$$

The geometrized Planck density is

$$\rho_p^{(geom)} = G\rho_p/c^2 = 1/\ell_p^2.$$

Equation (6.6) therefore becomes

$$a^3 = (3/4\pi)M\ell_p^2.$$

Equating the denominators of Eqs. (6.7) and (8.1),

$$2M\ell_\star^2 = \frac{3}{4\pi}M\ell_p^2$$

Hence

$$\ell_\star^2 = \frac{3}{8\pi}\ell_p^2$$

$$\ell_\star = \sqrt{\frac{3}{8\pi}}\ell_p$$

Under the minimal two-channel completion, the regularization scale is fixed by the Planck length rather than fitted independently for each compact object.

26. General p-channel completion and harmonic robustness

To test whether the exponent two is unique at the effective level, consider the generalized local closure

$$\rho/\rho_p = \chi^p.$$

For more than one multiplicative channel, the spherical mass equation can be integrated exactly. Using the asymptotic mass condition gives

$$m(r) = M1 + \left[\left(a^3/r^3 \right)^{p-1} \right]^{-1/(p-1)}$$

$$a^3 = 3M/(4\pi\rho_p).$$

All $p > 1$ solutions share the same Planck-density central limit,

$$m(r) \simeq (4\pi/3)\rho_p r^3 (r \rightarrow 0).$$

Thus Planck-density saturation by itself does not select the number of processing channels.

Why the single-channel case fails for an isolated finite mass

For a single multiplicative channel, the local density equals the mean enclosed density. The mass equation reduces to

$$dm/dr = 3m/r.$$

Its nonzero solution grows cubically with radius and therefore cannot approach a finite nonzero ADM mass at large radius. Within this local closure an isolated finite-mass solution needs more than one multiplicative channel; the minimal integer choice is therefore two.

Relation of the two-channel branch to the finite-Weyl completion

The finite-Weyl result of Section 5 does not make the effective channel-counting argument redundant; it supplies a second conditional route to the same quadratic response. The effective argument selects two as the smallest integer number of multiplicative channels compatible with finite isolated mass. The microscopic argument selects the first harmonic under the chord-load identification, positivity, saturation, and infrared-normalization assumptions. These are complementary rather than independent proofs. Full uniqueness remains open because the load dictionary, positivity restriction, and microscopic coarse graining remain additional assumptions.

Robustness to a second Weyl harmonic

As a controlled stress test, let

$$V(\delta) = a_1[1 - \cos(\delta)] + a_2[1 - \cos(2\delta)], \epsilon = a_2/a_1.$$

Using the chord-load definition, the normalized two-harmonic constitutive law becomes

$$\rho/\rho_p = f_\epsilon(\chi) = \chi^2[1 + 4\epsilon(1 - \chi^2)].$$

The saturated center remains exact. Positivity of the two-harmonic potential supplies the lower bound, while linear stability of the regular central branch supplies the upper bound. Together they define the consistency interval

$$-1/4 \leq \epsilon < 1/8.$$

At low load, the leading exterior correction remains quartic in inverse radius while its coefficient depends on microscopic harmonic content. The central Planck density and cube-root mass scaling are therefore more robust than the exterior correction coefficient.

27. Exterior corrections for general p

At large radius, Eq. (9.2) gives

$$m(r) \simeq M1 - [1/(p-1)](a^3/r^3)^{p-1} + \dots$$

In geometrized units, the first correction to the static metric function follows from the large-radius mass profile as

$$\Delta f(r) \propto r^{-(3p-2)}.$$

Therefore

$$p = 2 \Rightarrow \Delta f \propto r^{-4}$$

$$p = 3 \Rightarrow \Delta f \propto r^{-7}.$$

Higher-channel models are progressively more hidden from exterior observations. The two-channel case is simultaneously the minimal finite-mass integer closure and the least suppressed member of this regular family. Within the Weyl stress test, small higher-harmonic admixtures preserve the quartic inverse-radius power while modifying its coefficient.

28. Astrophysical consequences

For the minimal two-channel Planck-capacity completion, the characteristic core radius scales as

$$a \propto (M\ell_p^2)^{1/3}$$

whereas the Schwarzschild radius scales linearly with M . Hence

$$a/r_s \propto M^{-2/3}.$$

For stellar and supermassive black holes this ratio is extraordinarily small. The completion therefore does not naturally predict percent-level departures in ordinary shadow radii, ISCO locations, binary inspiral waveforms, or standard ringdown frequencies. Instead, the natural phenomenology is a general-relativistic exterior together with a finite central core.

The finite-Weyl bridge does not change this conclusion. In the preferred first-harmonic completion it reproduces the same Hayward-type profile. Higher-harmonic deformations can alter the coefficient of the leading quartic inverse-radius correction, but for a Planck-scale core the absolute exterior deviation remains extremely suppressed for astrophysical masses.

29. Horizons, echoes, and information

A regular center does not automatically eliminate an outer event horizon. It therefore does not, by itself, predict gravitational-wave echoes. Echoes require an additional reflecting or horizonless structure capable of returning information to the exterior.

Likewise, removal of the curvature singularity does not by itself solve the black-hole information problem. The information sector therefore has to be constructed independently of the finite core. Part IV develops such a construction: a rank-two outgoing boundary algebra is extracted from the strict-local four-component carrier, gauge-reduced horizon records are counted, an explicit finite-dimensional unitary evaporation map transfers those records to radiation/history, and a typical-state Page turnover is obtained. These results do not yet calculate exclusive microscopic amplitudes or the greybody spectrum numerically. Section 35 shows, however, that an independent horizon-source normalization is unnecessary at the inclusive level: microcanonical reciprocity ties spontaneous emission to absorption once the universal local QCA and exterior scattering problem are specified.

30. Finite-Weyl microscopic realization of the quadratic core law

Spherical load dictionary

$$\chi(r) = \frac{3m(r)}{4\pi\rho_p r^3}$$

For spherical strong gravity, define the load as the mean enclosed density in Planck-density units. The microscopic matching condition is

$$\chi_W(r) = \chi(r)$$

This identification is a completion choice. Under it, the positive-harmonic first-link response gives exactly

$$\rho(r)/\rho_p = \chi(r)^2$$

which is the minimal two-channel Compute-Link closure introduced above. The two interpretations are compatible: the quadratic response can be read both as the factorized support of two mandatory local sectors and as the squared finite-Weyl chord mismatch of the primitive Link record.

Hayward-type mass profile

$$\frac{dm}{dr} = 4\pi r^2 \rho_p \chi(r)^2$$

$$m(r) = \frac{Mr^3}{r^3 + \frac{3M}{4\pi\rho_p}}$$

$$a^3 = \frac{3M}{4\pi\rho_p}$$

The central density is finite and saturates at the Planck scale; the characteristic core size follows the cube-root mass law. At large radius, the leading static metric correction is quartic in inverse radius. The finite-Weyl construction therefore supplies a candidate microscopic realization of the closure introduced above as an explicit completion assumption.

Status of the derivation

The chain is conditional. Planck capacity fixes the saturation density and cube-root mass scale; finite information and Weyl kinematics provide the finite record carrier; positive-harmonic selection identifies the primitive cosine

interaction; and the chord-load dictionary maps that interaction to the coarse spherical load. The dictionary and harmonic positivity are not consequences of the capacity postulate alone and must remain separately testable assumptions.

31. Robustness to higher harmonics

Two-harmonic stress test

$$V(\delta) = a_1[1 - \cos(\delta)] + a_2[1 - \cos(2\delta)], \epsilon = a_2/a_1$$

Using the chord-load definition and normalizing by the first-harmonic saturation value gives

$$f_\epsilon(\chi) = \chi^2[1 + 4\epsilon(1 - \chi^2)]$$

The saturated center remains fixed. The spherical autonomous equation is

$$r \frac{d\chi}{dr} = 3[f_\epsilon(\chi) - \chi]$$

Linearization around the saturated center gives a regular inward approach over the upper-stability branch, while nonnegativity of the two-harmonic potential gives the lower bound. Together they define the allowed interval

$$-1/4 \leq \epsilon < 1/8$$

Robust versus model-dependent strong-gravity statements

At low load,

$$f_\epsilon(\chi) = (1 + 4\epsilon)\chi^2 - 4\epsilon\chi^4 + o(\chi^6)$$

Thus the leading exterior power remains quartic in inverse radius while its coefficient changes with the second harmonic. The Planck-density central limit and cube-root mass scaling are therefore more robust than the numerical coefficient of the far-field correction. This distinction is important for phenomenology: the existence and scaling of the core are capacity-level statements, whereas detailed exterior coefficients probe the microscopic harmonic content.

PART IV. BLACK-HOLE INFORMATION AND HORIZON-LINK ENTROPY

32. Why the Planck core cannot store the full black-hole entropy

$$N_{core} \sim M/m_p$$

$$S_{core}^{max} \sim k_B(M/m_p) \ln(d_c) \propto M$$

$$S_{BH} = \frac{k_B A_H}{4\ell_p^2} = 4\pi k_B \left(\frac{M}{m_p}\right)^2 \propto M^2$$

For any mass-independent finite local Hilbert-space dimension, the Planck-density core has parametrically too few independent states to account for the entropy of a large black hole. The new microscopic Link completion does not change this counting mismatch; it strengthens the distinction between core regularization and horizon-scale information accounting.

33. Horizon boundary algebra and the area law

Why raw graph crossings are not horizon qubits

The information-bearing horizon sector must be defined after physical coarse graining and gauge reduction, not by counting raw BCC edges. For a chosen surface normal, the raw crossing map obtained by summing outward BCC updates can have rank two for special orientations and rank four for generic orientations. Raw crossing-edge number and raw crossing rank are therefore regulator- and orientation-dependent. This rules out the naive identification “one graph edge crossing the horizon = one independent qubit.”

Orientation-independent normal-current sector

The doubled four-component Clifford carrier defines an orientation-independent normal-current split. For any unit physical normal, the Clifford relations force two outgoing and two ingoing eigenchannels. The corresponding projectors are

$$P_{out}(\hat{n}) = \frac{I+\alpha_n}{2}, \quad P_{in}(\hat{n}) = \frac{I-\alpha_n}{2}, \quad rank P_{out} = rank P_{in} = 2$$

Under a local frame rotation the normal-current operator changes by unitary conjugation, so the rank is gauge and basis independent. The effective outgoing boundary algebra is

$$A_H = P_{out} M_4(C) P_{out} \simeq M_2(C)$$

Its minimal finite Weyl algebra is the Pauli/Weyl algebra with a binary central phase label. Thus the strict-local bulk carrier supplies a natural two-state effective crossing sector rather than requiring a binary horizon record to be inserted by hand.

This kinematic result is not by itself a theorem about the full many-body horizon Hilbert space. The additional Boundary Record Minimality Principle used below states that, after the inside/outside split and gauge reduction, the minimum independent horizon record is precisely this outgoing support algebra, with no extra independent occupancy or species factor. Under that explicit assumption,

$$d_L = 2$$

Gauge-reduced state counting

Represent the effective horizon patches by binary boundary labels. For a closed connected horizon impose one global binary consistency condition:

$$\sum_{i=1}^F b_i \equiv 0 \pmod{2}$$

The physical state count is then

$$\Omega_F = 2^{F-1}, \quad S_H = k_B (F - 1) \ln 2$$

The missing factor of two is a single global consistency constraint and does not alter the macroscopic area density. In this sense, a central phase that is record-invisible in a closed bulk update becomes a physical boundary record when the system is cut by a horizon.

Area normalization by micro-IR matching

The finite-Weyl algebra fixes the information increment carried by one independent binary boundary record, but it does not by itself fix the corresponding physical area. That conversion is determined by the semiclassical GCC infrared limit for a stationary horizon:

$$\delta E = \frac{\kappa c^2}{8\pi G} \delta A, \quad k_B T_H = \frac{\hbar \kappa}{2\pi c}$$

Combining the first law with the Hawking temperature gives the Bekenstein-Hawking entropy-area conversion. Matching one independent boundary bit then yields

$$\Delta A = 4 \ln 2 \ell_p^2 \simeq 2.772588722 \ell_p^2$$

The Bekenstein-Hawking coefficient is therefore not claimed to follow from finite-Weyl kinematics alone. It follows from a micro-IR matching theorem: the microscopic boundary algebra fixes the entropy increment, while the independently required semiclassical GR/QFT limit fixes the conversion from information to physical area. The resulting effective patch is not a raw microscopic plaque.

34. Discrete horizon tower and unitary information transfer

Geometry and information labels

The boundary information labels should not be confused with geometry labels. At fixed horizon level n all binary boundary states share the same macroscopic geometry; geometry changes when the number of independent boundary records changes. The horizon Hilbert space is organized schematically as

$$H_H = \bigoplus_n |g_n\rangle \otimes H_{edge}^{(n)}, \quad \dim H_{edge}^{(n)} = 2^n$$

This removes the earlier temptation to assign a distinct geometry or area eigenvalue to the two values of an internal boundary label.

Discrete area and entropy tower

Using the regular extremal geometry as an information-free offset,

$$A_{ext} = \frac{9}{2} \ell_p^2 \quad r_{ext} = \frac{3\ell_p}{\sqrt{8\pi}}$$

Define

$$A_n = A_{ext} + 4n \ln 2 \ell_p^2$$

$$S_n = k_B n \ln 2 = \frac{k_B (A_n - A_{ext})}{4\ell_p^2}$$

Thus the endpoint may retain a finite geometric area while the boundary Hilbert space is one-dimensional. A geometric remnant is therefore not an information remnant: it need not store the information of the progenitor black hole.

Energy levels and finite gaps

For the regular spherical branch at the fixed Planck-scale core parameter, the horizon mass-radius relation can be written

$$M(r_h) = \frac{c^2}{G} \frac{r_h^3}{2(r_h^2 - \ell_*^2)}$$

Using the area level to define the horizon radius and inserting it into the mass-radius relation, the energy released in a one-record transition is

$$\epsilon_n = (M_n - M_{n-1})c^2$$

For n much greater than one the Schwarzschild limit gives

$$M_n \simeq m_P \sqrt{\frac{n \ln 2}{4\pi}}$$

$$\epsilon_n \simeq k_B T_H \ln 2$$

with the exact finite-step Schwarzschild ratio

$$\frac{\epsilon_n}{k_B T_H \ln 2} = 2n - 2\sqrt{n(n-1)} = 1 + \frac{1}{4n} + \frac{1}{8n^2} + O(n^{-3})$$

Explicit unitary evaporation map

Let the independent horizon records be binary. A one-step information-and-energy transfer can be written

$$|g_n; b_1, \dots, b_n\rangle |0\rangle_R \mapsto |g_{n-1}; b_1 \oplus b_n, b_2, \dots, b_{n-1}\rangle |b_n, \epsilon_n\rangle_R$$

where addition of the binary records is modulo two. The map is bijective: the emitted record is retained in radiation, and the original first record can be reconstructed from the updated first record and the emitted bit. It therefore extends to an exact finite-dimensional unitary. Energy conservation is blockwise by construction,

$$M_n c^2 = M_{n-1} c^2 + \epsilon_n$$

Repeated application transfers all progenitor boundary information to radiation/history correlations without requiring the final geometric endpoint to carry a large hidden Hilbert space.

Near-endpoint thermodynamic work term

For the regular branch, applying the ordinary first law to the entire static family while holding the core scale fixed ceases to be exact near the core. The discrepancy vanishes in the Schwarzschild limit and at the cold extremal endpoint and is naturally represented as

$$dE = T dS + dW_{core/link}$$

Accordingly, the constant information-area spacing is interpreted as the boundary information calibration, while the near-core static family can exchange additional work with the core/Link sector. This prevents the regular endpoint from being overinterpreted as an ordinary Schwarzschild thermodynamic state.

35. Hawking detailed balance, evaporation rate, and Page turnover

State-counting origin of the thermal factor

Allow a transition that removes q boundary records. Its exact geometry energy is

$$\epsilon_{n,q} = (M_n - M_{n-q})c^2$$

Because the number of boundary states doubles for each independent record, microreversibility gives the coarse-grained state-count ratio

$$\frac{\Gamma_{emit}}{\Gamma_{absorb}} = \frac{\Omega_{n-q}}{\Omega_n} = 2^{-q}$$

For a macroscopic Schwarzschild black hole,

$$\epsilon_{n,q} \simeq q k_B T_H \ln 2$$

so

$$2^{-q} \simeq \exp\left[-\frac{\epsilon_{n,q}}{k_B T_H}\right]$$

The Hawking Boltzmann factor is therefore compatible with boundary-state counting rather than being inserted as an independent microscopic probability. At finite n the gap is slightly nonthermal:

$$\frac{\epsilon_{n,q}}{q k_B T_H \ln 2} = \frac{2n}{q} \left(1 - \sqrt{1 - \frac{q}{n}}\right) = 1 + \frac{q}{4n} + \frac{q^2}{8n^2} + O(n^{-3})$$

Horizon-blindness and no independent surface coupling

In GCC the primitive update is universal and strict-local: the gate acting on a neighborhood may depend on local geometry and matter records, but it is not supplied with a separate global label stating that the neighborhood lies on an event horizon. This matters because event-horizon membership is a global causal property, while the local curvature at the horizon of a sufficiently massive black hole can be arbitrarily small. A special surface coupling that survived in that limit would require an additional nonlocal horizon flag and would violate the locality/no-retuning interpretation of the microscopic law. The minimal branch therefore imposes no boundary defect. Reflection and transmission seen from infinity arise from propagation through the exterior geometry. Denote the resulting inclusive channel absorption probability by the greybody function used below.

Microcanonical reciprocity and inclusive normalization

Let the transition amplitude connect an initial black-hole microstate in the lower horizon sector plus one incoming asymptotic quantum to a final microstate in the higher sector. Microscopic reversibility equates the modulus of this amplitude with that of the inverse emission channel. Define the microcanonical averages

$$P_{abs}(n - q \rightarrow n; \alpha) = \frac{1}{\Omega_{n-q}} \sum_{i,j} \left| M_{ji}^{(\alpha)} \right|^2, \quad P_{emit}(n \rightarrow n - q; \alpha) = \frac{1}{\Omega_n} \sum_{j,i} \left| M_{ji}^{(\alpha)} \right|^2$$

The same matrix-element sum appears in both expressions, so the unknown exclusive amplitudes cancel in the ratio:

$$\frac{P_{emit}}{P_{abs}} = \frac{\Omega_{n-q}}{\Omega_n} = 2^{-q}$$

For the no-boundary-defect branch, the absorption probability is precisely the inclusive QCA greybody probability at the relevant transition frequency. Hence

$$P_{emit}(n \rightarrow n - q; \alpha) = 2^{-q} T_\alpha(\omega_{n,q})$$

This is stronger than detailed balance alone: once the absorption problem is calculated, there is no independent horizon-source normalization to fit.

Bose/Fermi occupation and QCA flux normalization

For a general emitted quantum, define the black-hole entropy loss as the difference between the initial and final black-hole entropies. Repeated occupation of a bosonic mode gives a geometric series, while Pauli exclusion truncates fermionic occupation. In the smooth continuum normalization the inclusive number flux therefore takes the form

$$\frac{d^2 N}{dt d\omega} = \frac{1}{2\pi} \sum_{\alpha} \frac{T_\alpha(\omega)}{\exp[\Delta S_{BH}(E, \omega)/k_B] \mp 1}$$

with the minus sign for bosons and the plus sign for fermions. In the macroscopic limit the black-hole entropy loss becomes the emitted quantum energy divided by the Hawking temperature, recovering the usual greybody form.

For the exact QCA, the continuum radial measure is replaced when necessary by the density-of-states and group-velocity measure implied by the QCA dispersion and physical tick. No new emission constant is introduced.

Sector-centroid line structure and microstate splitting

The discrete area tower implies discrete transition frequencies,

$$\hbar\omega_{n,q} = \epsilon_{n,q}$$

If every microstate inside a fixed horizon sector were exactly degenerate at its sector mass, the minimum model would predict a line spectrum with a thermal envelope rather than an exactly continuous Planck spectrum. Exact degeneracy is not required. A full interacting QCA may instead split the sector into exponentially many nearby levels while preserving the total state count. The centroid spacing remains set by one boundary-record entropy unit, while individual transition frequencies can become dense. Resolvable Hawking lines are therefore a conditional prediction of a weak-splitting branch.

Exact per-tick emission dilation

For a chosen emission channel, an exact one-tick unitary can be defined on the paired no-emission/emission subspace as a two-dimensional rotation,

$$|n, \mu; 0\rangle \mapsto \sqrt{1-p} |n, \mu; 0\rangle + \sqrt{p} V |n, \mu\rangle, \quad V |n, \mu\rangle \mapsto -\sqrt{p} |n, \mu; 0\rangle + \sqrt{1-p} V |n, \mu\rangle$$

Fresh radiation ancillas over repeated ticks give an exponential waiting-time law. For an exclusive microstate channel the detailed per-tick probability still requires the full interacting QCA. However, after microcanonical averaging it is not an independent phenomenological horizon parameter: the summed emission probability is fixed by the inverse absorption problem and the horizon-state degeneracy ratio derived below.

Macroscopic inclusive rate and scaling

After the absorption problem is specified, the inclusive power is determined without an independent luminosity coefficient. Let the asymptotic mode-flux density be the one fixed by the QCA dispersion and physical tick. The microcanonical expression is

$$P(E) = \sum_{\alpha} \int d\omega \frac{\hbar\omega \Phi_{QCA,\alpha}(\omega) T_{\alpha}(\omega)}{\exp[\Delta S_{BH}(E,\omega)/k_B] \mp 1}$$

In the continuum radial normalization this flux measure reduces to the familiar partial-wave measure; the exact finite-QCA expression instead uses the corresponding group-velocity and density-of-states Jacobian.

For a macroscopic Schwarzschild black hole, the Hawking temperature carries one inverse power of mass, and the exterior scattering problem depends only on dimensionless frequency combinations. Rescaling to that dimensionless frequency gives inverse-square luminosity and cubic lifetime. The same exponents follow independently from GCC causality: the collective attempt frequency is inverse in mass, and the typical one-record energy is also inverse in mass. Treating microscopic patches as independently emitting at the Planck rate would give the wrong scaling and is excluded.

Page turnover

For an initially scrambled horizon, after a given number of records have been transferred to radiation, the typical pure-state radiation entropy is

$$S_R(n) \simeq k_B \ln 2 \min(n_0 - n, n)$$

The Page turnover occurs when half the original boundary-record capacity remains. In the macroscopic area-mass relation this implies

$$M_{Page} = \frac{M_0}{\sqrt{2}}$$

Using the simplified Schwarzschild mass-loss law then gives

$$\frac{t_{Page}}{\tau} = 1 - \frac{1}{2\sqrt{2}} \simeq 0.6464466$$

Thus “half the horizon information capacity” does not correspond to half of the physical evaporation time.

Matter coupling and remaining exclusive dynamics

Minimal metric coupling still controls the exclusive geometry-to-matter transition structure. Its schematic variation is

$$\delta S_m = \frac{1}{2} \int d^4x \sqrt{-g} T^{\mu\nu} \delta g_{\mu\nu}$$

For free fields the stress tensor is quadratic, so the exclusive fundamental creation process is generically pair/squeezing-like rather than a literal one-particle lowering operator. A full nonlinear geometry/matter QCA is still required to determine individual microstate amplitudes, species and polarization branching, level splittings, and linewidths. Their common overall normalization, however, is not a separate horizon parameter at the inclusive level: the summed matrix-element strength is fixed by the inverse absorption probability through microcanonical reciprocity. The remaining greybody factors are scattering observables of the same exterior QCA/geometry, not adjustable Hawking coefficients.

PART V. CLAIMS, LIMITATIONS, AND FALSIFIABILITY

36. Claim-status summary

Element	Status	Meaning
Finite local information	Explicit postulate	Finite-dimensional local quantum records.
Finite Weyl/symplectic carrier	Preferred candidate architecture	Exact finite microscopic unitary composition.
Microscopic closure	Exact at update/groupoid level	The ultraviolet consistency statement is finite, not continuum-infinitesimal.
ADM/DeWitt constraint algebra	Emergent infrared target	Recovered in smooth/refinement sectors; full-cutoff exact HDA is not claimed.
Planck-density regular core	Robust capacity result	The density cap and cube-root core scaling follow before choosing the detailed interpolation.
Minimal two-channel quadratic profile	Conditional completion	Requires the stated Compute-Link/chord-load assumptions.
Horizon boundary entropy coefficient	Conditional micro-IR matching	Rank-two boundary algebra selects a binary boundary record; the area-law coefficient follows from semiclassical micro-IR matching, conditional on Boundary Record Minimality.
Full 3+1 many-body dynamics	Open	A strict-local linear propagation QCA is explicit; a unique full nonlinear geometry/matter update and physical-state construction remain to be derived.

37. Falsifiability and consistency tests

- Infrared coarse graining must recover the complete ADM/DeWitt H-H, H-D, and D-D algebra, not only one bracket.
- The finite microscopic carrier must possess a nonempty physical Hilbert space after all constraints and gauge reductions are imposed.
- Any ultraviolet deformation used as new physics must preserve a first-class physical constraint algebra or be demonstrably pure gauge; isolated term-by-term polymerization is insufficient.
- The record-preserving linear scheduler must remain stable under graph refinement and must not generate uncontrolled macroscopic Lorentz or diffeomorphism violation.
- The Weyl chord-load dictionary must be derived, independently motivated, or empirically constrained; otherwise the microscopic link to the minimal two-channel regular-core completion is lost.
- The positive-harmonic assumption must be justified by stability or microscopic energy positivity. If negative harmonics are physically required, the first-harmonic uniqueness proposition no longer holds.

- The spherical strong-gravity sector must respect the Planck-density bound and recover cube-root mass scaling. Detailed exterior coefficients should be treated as microscopic-model dependent until the harmonic content is fixed.
- The explicit reversible Compute-Link correlation circuit must survive extension to the full finite-dimensional many-body geometry sector; its correlation profile must not be replaced by an arbitrary radial interpolation.
- The rank-two horizon boundary algebra and binary gauge reduction must survive a full many-body construction. Boundary Record Minimality, rather than a fitted state count, is the remaining microscopic assumption behind the two-state boundary record.
- The explicit unitary evaporation map must be embedded in the full matter/radiation QCA, preserve energy under the microscopic update law, and transfer progenitor information without a large-information remnant.

38. Strong-gravity limitations and falsification conditions

The present construction does not establish the following:

- a first-principles derivation of the Weyl chord-load dictionary;
- microscopic uniqueness beyond the explicitly stated positive-harmonic sector;
- equality of the Compute and Link load fractions in a generic ultraviolet quantum state;
- dynamical formation of the static profile during collapse;
- inner-horizon stability;
- a rotating Kerr-like completion;
- a completed numerical greybody/channel spectrum, exclusive microstate transition amplitudes and linewidths, and a dynamical proof of endpoint stability;
- a full interacting-field Page curve beyond the typical-state finite-record model;
- a derivation showing that the full nonlinear finite-Weyl scheduler coarse-grains uniquely to the static spherical constitutive law used here.

Most importantly, Planck capacity alone does not uniquely imply the Hayward profile. The Hayward form follows from Planck capacity plus the minimal two-channel local constitutive law. The finite-Weyl Link construction supplies a conditional microscopic realization and a positive-harmonic uniqueness proposition within a restricted sector, but negative harmonic coefficients, alternative load dictionaries, or more general interactions can evade that argument. The capacity-derived density bound and cube-root mass scaling remain the more robust results.

The completion is falsified as a physical model if its required microscopic dynamics cannot preserve the successful low-energy GR correspondence, if the saturated core is dynamically unstable, or if a consistent rotating/evaporating extension cannot be constructed without violating the finite-capacity and no-retuning principles.

39. Interpretation as a quantum-gravity candidate

The architecture obtained here has three attractive features. First, the microscopic carrier is genuinely finite and unitary. Second, the metric-dependent H-H structure function is generated geometrically from the response of the normal rather than added as an arbitrary ultraviolet coefficient. Third, the failure of exact continuum derivations at finite cutoff is not hidden; it is isolated and converted into a controlled emergence requirement.

These features are not sufficient to establish a complete theory of quantum gravity. The current result is best described as a finite-information quantum-gravity candidate architecture with exact finite microscopic composition, an anomaly-free emergent HDA in the tested smooth sectors, a strict-local BCC propagation candidate, a rank-2 horizon boundary algebra, and an explicit unitary black-hole information-transfer map.

40. Open problems and decisive next tests

- Extend the strict-local four-component BCC propagation QCA to one nonlinear geometry/matter many-body update on the full tensor-product graph Hilbert space.
- Demonstrate the complete 3+1 Lorentzian constraint and physical-state structure, including the role of local frame/Gauss gauge in the chosen microscopic variables.

- Compute the absorption/greybody matrix of the strict-local matter QCA on the regular GCC exterior and use microcanonical reciprocity to obtain the inclusive emission spectrum; derive exclusive pair/squeezing amplitudes from the full nonlinear geometry/matter update.
- Define a controlled refinement/renormalization map connecting finite d and finite graph spacing to the continuum observables.
- Show that the finite-Weyl strong-gravity response, regular core, constraint closure, horizon boundary algebra, and evaporation source arise from one common strict-local microscopic update rule.
- Construct physical states or an appropriate groupoid-averaging analogue and demonstrate unitary relational evolution.
- Derive Boundary Record Minimality or an equivalent gauge-reduction theorem, and determine the microstate level splittings and linewidths. The two-state boundary algebra, micro-IR area matching, unitary transfer map, microcanonical emission/absorption normalization, and typical-state Page turnover are explicit candidate results.
- Compute the microstate energy splittings and linewidths to decide whether the sector-centroid spacing produces resolvable Hawking lines or a dense continuum-like spectrum; retain the finite-n nonthermal entropy correction as the more robust target.

The candidate is falsified or requires major revision if no single microscopic update law can satisfy these conditions simultaneously, if refinement fails to recover the ADM/DeWitt universality class, if the finite-capacity strong-gravity branch is dynamically inconsistent, or if matter coupling requires incompatible causal/update structures.

41. Integrated conclusion

The construction developed in this paper can be summarized in four levels. First, the microscopic substrate is finite: bounded energy capacity and finite information are distinct, and the preferred record algebra is finite Weyl/symplectic. Exact consistency is assigned to finite update composition. A strict one-neighborhood-per-update BCC QCA provides a concrete linear propagation carrier with two positive-frequency amplitudes; the primitive law is the QCA itself, while a Hamiltonian is only an infrared effective description. Second, smooth graph sectors recover geometry and the ADM/DeWitt deformation algebra. The inverse-metric H-H structure is generated by the state-dependent normal, while H-D and D-D recover their continuum forms under refinement. Third, high gravitational load is bounded. The Planck-density cap gives the characteristic cube-root mass scaling of the core, and the quadratic Compute-Link/chord-load completion produces a Hayward-type nonsingular profile. Fourth, the horizon information sector is separated from the core: the rank-two normal-current algebra supplies a binary effective boundary record, semiclassical matching fixes its physical area conversion, and the resulting horizon tower supports exact energy-conserving unitary information transfer.

$$A_{micro} = 0, \quad A_{eff} \rightarrow 0$$

$$\rho \leq \rho_p, \quad r_c \propto M^{1/3}, \quad m(r) = \frac{Mr^3}{r^3 + 3M/(4\pi\rho_p)}$$

The final mass profile displayed above is the conditional result of the minimal quadratic completion. The framework is not presented as a finished unique quantum theory of gravity. The remaining gaps are now more localized: the strict-local propagation QCA must be extended to the full nonlinear geometry/matter Hilbert space; Boundary Record Minimality must be derived or independently justified; and the absorption/greybody spectrum, exclusive microstate amplitudes, level splittings and linewidths must be calculated from that same update law. An independent phenomenological horizon-source constant is not required at the inclusive level, because microcanonical reciprocity relates emission to absorption through the horizon-state degeneracy ratio. Subject to these conditions, the paper now provides more than a static regular-core model: it contains a candidate microscopic boundary algebra, a sector-centroid horizon spectrum, a Hawking detailed-balance and normalization mechanism, causal evaporation scalings, and a typical-state Page turnover. These are concrete falsification targets rather than placeholders for future information-loss dynamics.

APPENDIX A. General p-channel derivation

Starting from the general multiplicative-channel constitutive law and the mean-density load variable, the mass equation is

$$dm/dr = 3^p (4\pi\rho_p)^{1-p} m^p r^{2-3p}. \quad (A. 1)$$

For any channel number other than one, separation and integration give

$$m^{1-p} = C + \left[3/(4\pi\rho_p)\right]^{p-1} r^{-3(p-1)}. \quad (A.2)$$

For more than one channel, the asymptotic mass condition fixes the integration constant. Factoring the result and defining the characteristic core scale yields Eq. (9.2). The single-channel case must be solved separately and gives cubic mass growth.

APPENDIX B. Positive-harmonic Weyl selection

Let the Link potential be expanded in nonnegative cosine harmonics and define the normalized chord load. Full phase separation fixes the saturation value. Near the vacuum, the phase is twice the chord load to leading order, so

$$V(\delta) = 2\chi_w^2 \sum_m m^2 a_m + O(\chi_w^4)$$

After normalizing by the saturation value, exact unit infrared coefficient requires Eq. (5.9). The difference between the two sides is

$$\sum_{m \text{ even}} m^2 a_m + \sum_{m \text{ odd } m>1} (m^2 - 1) a_m$$

Every term in this expression is nonnegative, so equality forces all coefficients except the first harmonic to vanish. The result is conditional on the nonnegative-harmonic assumption and the stated saturation and infrared normalization.

APPENDIX C. Numerical audit definitions

A.1 Weyl audit

For the finite-Weyl audit, the cyclic shift and clock matrices were constructed directly. The reported residual uses one pair of low-order Weyl words, with the phase exponent evaluated modulo the local Hilbert-space dimension.

A.2 One-dimensional finite-difference audit

The periodic interval uses the standard Fourier normalization, and the centered derivative is the standard second-order stencil. The smooth test fields are

$$N(x) = 0.8 + 0.25\sin x + 0.10\cos(2x)$$

$$v(x) = 0.6 + 0.20\cos x - 0.08\sin(2x)$$

$$w(x) = -0.3 + 0.22\sin x + 0.12\cos(3x)$$

With the lapse multiplication and tangential transport operators defined as above, the residuals are

$$R_{HD} = [L_v, M_N] - \text{diag}(v D_a N)$$

$$R_{DD} = [L_v, L_w] - \text{diag}(v D_a w - w D_a v) D_a$$

Relative Frobenius norms are reported. Smooth-sector norms keep only the fixed low Fourier band while the graph is refined.

A.3 Three-dimensional normal-refresh audit

The graph audit uses centered derivatives on the parameter cube and reconstructs a unit normal in four-dimensional embedding space from the three tangent records. Two outer layers are removed from norm estimates. After each normal update the tangents and normal are recomputed from the updated embedding. The fitted coefficient is the least-squares projection of the finite-path order difference onto the continuum tangential-shift predictor given in Section 12.

APPENDIX D. Logical reading of exactness

Three notions of exactness should not be conflated. (i) Matrix exactness: finite Weyl and finite gauge operators obey exact matrix identities. (ii) Microscopic path consistency: finite scheduler compositions differ only by admitted finite gauge/address arrows. (iii) Continuum constraint exactness: the ADM/DeWitt HDA holds as an infinitesimal algebra. This framework assigns (i)-(ii) to the finite microscopic theory and requires (iii) only in the appropriate emergent smooth sector.

APPENDIX E. Compact architecture audit

Layer	Requirement	Current status
Microscopic kinematics	Finite local Hilbert spaces and bounded observables	Specified
Microscopic algebra	Finite Weyl and symplectic/Clifford carrier	Specified at candidate level
Microscopic closure	Exact finite update/groupoid composition	Established as design principle; exact in finite Weyl sector
Emergent geometry	Metric/triad records from finite relational data	Proof-of-principle
Infrared gravity	ADM/DeWitt HDA in smooth sector	Analytic target + numerical pilot support
Ultraviolet strong gravity	Finite capacity and Planck-density saturation	Derived at effective level
Regular black hole	Quadratic load law and Hayward-type profile	Derived conditionally
Quantum information	Rank-two horizon algebra and unitary information transfer	Conditional candidate
Full dynamics	Unique 3+1 many-body Lorentzian update	Open
Phenomenology	Sector-centroid gaps, microstate splitting, and finite-sector corrections	Conditional spectral prediction

APPENDIX F. Horizon counting and area-normalization audit

F.1 Finite geometry record and area spectrum

The horizon information count does not require each raw graph face to carry an integer multiple of Planck area. If a finite inverse-triad record defines the positive metric matrix, the physical area of a coordinate face is schematically

$$A_n(E) = \ell_P^2 \sqrt{\frac{n_a G^{ab} n_b}{\det G}}$$

A finite geometry sector therefore gives a discrete area spectrum, but finite-Weyl kinematics alone does not fix the dimensionless spacing of that spectrum. Different admissible finite record maps can preserve the same infrared metric architecture while changing microscopic area eigenvalues.

F.2 Transfer-matrix formulation

For gauge-reduced boundary types, let the integer matrix count admissible gauge-inequivalent extensions and let the corresponding dimensionless matrix encode area increments. Define

$$(T_x)_{\alpha\beta} = C_{\alpha\beta} \exp[-x a_{\alpha\beta}], \quad \rho(T_x) = 1, \quad x_* = \frac{1}{4} \Rightarrow \rho(T_{1/4}) = 1$$

At large area the state count grows with the Perron exponent defined by the transfer matrix. The Bekenstein-Hawking density corresponds to the one-quarter value of that exponent.

F.3 Uniform-area finite-state no-go

$$x_* = \frac{\ln \rho(C)}{a_0}, \quad x_* = \frac{1}{4} \Rightarrow \rho(C) = e^{a_0/4}$$

If every allowed step has the same nonzero algebraic area and the transition grammar is a finite nonnegative integer matrix, the Perron eigenvalue is algebraic. By contrast, the exponential on the right-hand side is transcendental for nonzero algebraic area. A finite integer grammar with uniform algebraic microscopic area therefore cannot by itself produce the exact one-quarter coefficient. This is why the present construction separates microscopic information counting from the infrared conversion to physical area.

Part IV therefore uses two logically distinct ingredients: the rank-two boundary algebra fixes the entropy increment of one independent binary record, while semiclassical horizon thermodynamics fixes the physical area carried by

that increment. Nonuniform metric-weighted area spectra, refinement limits, or renormalized effective boundary grammars remain possible alternatives but are not required for the minimal branch developed here.

References

- [1] S. A. Hojman, K. Kuchar, and C. Teitelboim, “Geometrodynamics Regained,” *Annals of Physics* 96, 88-135 (1976).
- [2] S. A. Hayward, “Formation and evaporation of nonsingular black holes,” *Physical Review Letters* 96, 031103 (2006).
- [3] M. Visser, “Physical observability of horizons,” *Physical Review D* 90, 127502 (2014).
- [4] M. Simpson and M. Visser, “Regular black holes with asymptotically Minkowski cores,” *Universe* 6, 8 (2020).
- [5] C. Blohmann, M. C. B. Fernandes, and A. Weinstein, “Groupoid Symmetry and Constraints in General Relativity,” *Communications in Contemporary Mathematics* 15, 1250061 (2013).
- [6] B. Bahr and B. Dittrich, “Breaking and Restoring of Diffeomorphism Symmetry in Discrete Gravity,” *AIP Conference Proceedings* 1196, 10-17 (2009).
- [7] B. Bahr and B. Dittrich, “Improved and Perfect Actions in Discrete Gravity,” *Physical Review D* 80, 124030 (2009).
- [8] V. Bonzom and B. Dittrich, “Dirac’s Discrete Hypersurface Deformation Algebras,” *Classical and Quantum Gravity* 30, 205013 (2013).
- [9] T. Thiemann, “Exact Quantisation of $U(1)^3$ Quantum Gravity via Exponentiation of the Hypersurface Deformation Algebroid,” *Classical and Quantum Gravity* 40, 245003 (2023).
- [10] G. M. D’Ariano, M. Erba, and P. Perinotti, “Isotropic quantum walks on lattices and the Weyl equation,” *arXiv:1708.00826* (2017).
- [11] P. Raynal, “Simple derivation of the Weyl and Dirac quantum cellular automata,” *Physical Review A* 95, 062344 (2017), *arXiv:1703.05890*.
- [12] S. W. Hawking, “Particle Creation by Black Holes,” *Communications in Mathematical Physics* 43, 199–220 (1975).
- [13] D. N. Page, “Particle emission rates from a black hole: Massless particles from an uncharged, nonrotating hole,” *Physical Review D* 13, 198–206 (1976).
- [14] R. Emparan and I. Sachs, “Quantization of AdS_3 Black Holes in External Fields: Semiclassical Results from Pure Gravity,” *Physical Review Letters* 81, 2408–2411 (1998).